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Reaching carbon neutrality in China: Temporal and subnational limitations of renewable energy scale-up

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ABSTRACT

Acute temporal impacts and subnational limitations can hinder a country's decarbonization pathway, despite national planning efforts. China, as the world's largest greenhouse gas (GHG) emitter, has announced an ambitious climate policy goal of achieving carbon neutrality by 2060, which will require an unprecedented scale-up of low-carbon energy technologies. China's variable renewable energy (VRE) deployment is historically imbalanced with large geographic concentrations driven by resource endowment and institutional heterogeneities. If continued, this pattern can run into deployment limits and exacerbate challenges associated with socio-economic benefits distribution, threatening the ability to timely integrate VRE. We develop a capacity expansion model with grid operational detail and high spatial resolution to examine decadal pathways to carbon neutrality by 2060 considering localized and temporal impacts. Over these four decades, we find that all regions will increase deployment rates of renewable energy, first driven by the use of high-quality resources, and later by coal retirement and electricity demand growth. The share of provinces with high deployment pressure, where deployment requirements exceed historical rates, increases from around 45% to 100% by the final decade. If carbon capture and storage (CCS) is not available, maximum annual average deployment rates will increase by 33% and occur a decade earlier. A more stringent 1.5 °C emission target leads to more acute temporal and spatial deployment pressures in the first decade, with VRE concentrated in regions with high-quality resources and demand centers and a doubling of new transmission capacity in the first decade. Effective national and subnational policy support is necessary to coordinate VRE deployment and facilitate transitions in impacted regions.

1. Introduction

China is the world's largest greenhouse gas (GHG) emitter and has announced ambitious climate policy goals of reaching peak carbon emissions by 2030 and carbon neutrality by 2060 [1]. To achieve these goals, it is crucial to decarbonize the largest carbon-emitting source, the power sector, which further enables the electrification of other sectors such as transportation, industry, and buildings [2,3]. This process entails a large increase in low-carbon renewable energy and complementary infrastructure including storage and transmission. Despite these clear long-term directions, there is uncertainty about the pace, structure, and distributional impacts of the transition to a net-zero future, requiring models with sufficient granularity to uncover feasible and efficient pathways to carbon neutrality for China's power sector [4–7].

China has a large land area and high regional economic and ecological variability. Chinese regions differ in their urbanization trends, energy use, and availability of high-quality renewable resources. Geographically, over 60% of the installed wind and solar capacities are in China's Three-North regions that are endowed with high-quality renewable resources but only cover 40% of electricity demand [8]. This discrepancy has implications for socio-economic impacts of China's energy transition and calls for spatial and temporal modeling granularity to understand the variation and trade-offs in deployment, technology choices, and land use decisions to meet short- and long-term decarbonization goals. If the uneven pattern persists, it may exacerbate several issues, including those related to physical limits [9], welfare distributions [10], and societal benefits [11]. These can apply to upstream sectors (i.e., manufacturing, minerals, and materials) [12] as

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well as downstream sectors (i.e., deployment, power, and substitute sectors such as coal) [7]. These political economy factors may constrain deployment at the local level despite national planning efforts and are distinct from other well-studied techno-economic barriers such as renewable energy costs [7] and grid connection [13,14]. Lastly, as regions grapple with workforce displacement and stranded infrastructure associated with the low-carbon transition, equity concerns could lead to policy softening and pushback [10,15].

1.1. Literature review and research gap

In this context, we address the critical research question: how might spatial–temporal variations in VRE deployment affect feasible transition pathways to carbon neutrality? To frame our contribution, we position our work with respect to three categories of existing literature: (1) integrated assessment models (IAMs), (2) single-period power systems optimization models (snapshots), and (3) multi-period power systems optimization models (pathways). Our synthesis of the existing literature on modeling continental-scale power systems highlights a key research gap: a lack of integration between grid operational and land use constraints, with the analysis of acute renewable deployment pressure at subnational levels.

Studies using IAMs are valuable in providing economy-wide multi-decadal decarbonization frameworks. However, a significant limitation is the lack of spatial resolution and grid operational details to explore localized impacts of transition patterns. This structural gap leads to large variations in renewable energy deployment requirements across IAM-based studies. For example, Duan et al. conducted a multimodal comparative analysis and found that for China to meet the 1.5 °C target by 2060, renewable energy capacities vary by an order of magnitude, ranging from 1 terawatt (TW) to 11 TW [2]. Similarly, Kong et al. assessed 67 carbon neutrality scenarios for China's power sector and identified significant disparities in the deployment scale of solar energy by up to 160% based on different cost expectations about low-carbon technologies [16]. Despite the broad consensus that wind and solar will dominate China's energy and power sectors by mid-century [2,3,17], there is an inconsistency regarding the timing of accelerating renewable deployment paces, such as near-term urgency in the 2020s [2] and emphasis in the 2030s [3]. The uncertainty in transition pathways obtained from IAMs can be traced back to technology representations and, for the case of renewable energy, spatio-temporal granularity [16,17].

Engineering-economic optimization models can examine the power sector, either as snapshots in future years or pathways towards mid-century. Snapshot analyses typically specify an end-state and thus make strong assumptions about greenfield investments and do not explicitly incorporate intertemporal decisions of energy infrastructures. For instance, Guo et al. developed a capacity expansion model for China's power system and found that offshore wind capacity could reach 1500 GW by 2050, though substantial implementation barriers remain to expand this sector by an order of magnitude [14]. Similarly, Brown et al. examined how inter-regional coordination and transmission expansion reduce system costs for a 100% decarbonized U.S. electricity system, but emphasized real-world challenges for transmission construction, which requires overcoming barriers over time [18]. Many other studies have explored the potential of individual technologies in global power systems, including offshore wind [19–21], and distributed solar [22,23]. However, the modeling approaches differ depending on the examined technology and the assumptions made about interactions with other generation technologies. To enable a more comprehensive analysis, we consider the spatio-temporal variability of all major clean technologies within a unified modeling framework.

In comparison, pathways studies using power systems optimization models reveal the intertemporal dynamics of low-carbon transition over multiple time periods, such as changes in electricity costs [6], impacts of renewable cost declines [4,24], interactions of technology availability [5], and emission peaking timelines [25]. Approaches range from perfect foresight models, which assume full information of future uncertainties, to sequential myopic models, which reflect more realistic decision-making based on limited knowledge and evolving conditions [26–28]. While these studies capture varying degrees of techno-economic feasibility, they exhibit significant variations in key modeling features, resulting in a wide range of uncertainty in modeling outcomes. This highlights the need for continued methodological and analytical improvements. First, as the VRE penetration increases, diurnal and seasonal balancing requirements will be more pronounced. Capturing these dynamics necessitates higher temporal resolution, often requiring full-year, hourly modeling of grid operational and resource profiles [6, 29–31]. Second, coarser spatial resolution and the use of clustering techniques for candidate renewable energy sites can substantially affect estimates of transmission expansion, energy storage, and firm capacity needs [4–6,18,30,32]. Third, integrating renewables into the grid requires expansion of both inter-provincial and intra-provincial lines. Ignoring intra-provincial connection costs or relying on ex-post adjustments can lead to structural shifts across technology types and geographical locations [6,14]. Lastly, local political economy challenges and temporal limitations that arise during large-scale deployment, can constrain renewable and infrastructure deployment at subnational levels. However, since most models lack sufficient spatio-temporal granularity to capture these dynamics, there is limited analysis of provincial deployment bottlenecks and regional coordination efforts required in the face of technology availability and renewable cost uncertainties [29,33,34].

1.2. Study contributions and overview

Our study has both methodological advancements and analytical contributions to researchers in this field and policymakers, listed as follows:

1. Spatio-temporal granularity of renewable resources: we develop a sequential capacity expansion planning model with high spatial–temporal resolutions. This allows us to capture renewable resource profiles and land use details, and obtain more accurate estimates of resource potential and quality. Compared to other IAMs and optimization-based power sector modeling studies, our approach includes over 10,000 grid cells for renewable energy rather than typical single or clustered subregional profiles. We also model hourly grid operations and resource profiles across 8760 h of the year instead of using representative days or weeks, while accounting for granular representations of connection costs.
2. Build-out requirements over time and by subnational location: we quantify the temporally acute deployment pressure at subnational levels under uncertain factors. This allows us to identify region-specific policy priorities and coordination mechanisms for local governments.
3. Implications for China's decarbonization pathways: we apply the model to China's power system and reveal strong geographical disparities in renewable deployment, infrastructure needs, and deployment feasibility over time. These results offer practical guidance for aligning national policy with province-specific implementation capacity.

We develop a novel modeling framework with high spatial and temporal resolutions to identify pathways for deploying renewables, storage systems, and transmission lines, and retrofitting or retiring thermal generators, by decade, from 2020 to 2060 (see Fig. 1 and methods). Each decadal run uses the Renewable Energy Siting and Power-system Optimization (RESPO) model to optimize for renewable deployment and other variables as a snapshot [35]. We extend the RESPO model by endogenizing firm power deployment decisions and using decadal results from one run as the lower bound for the next

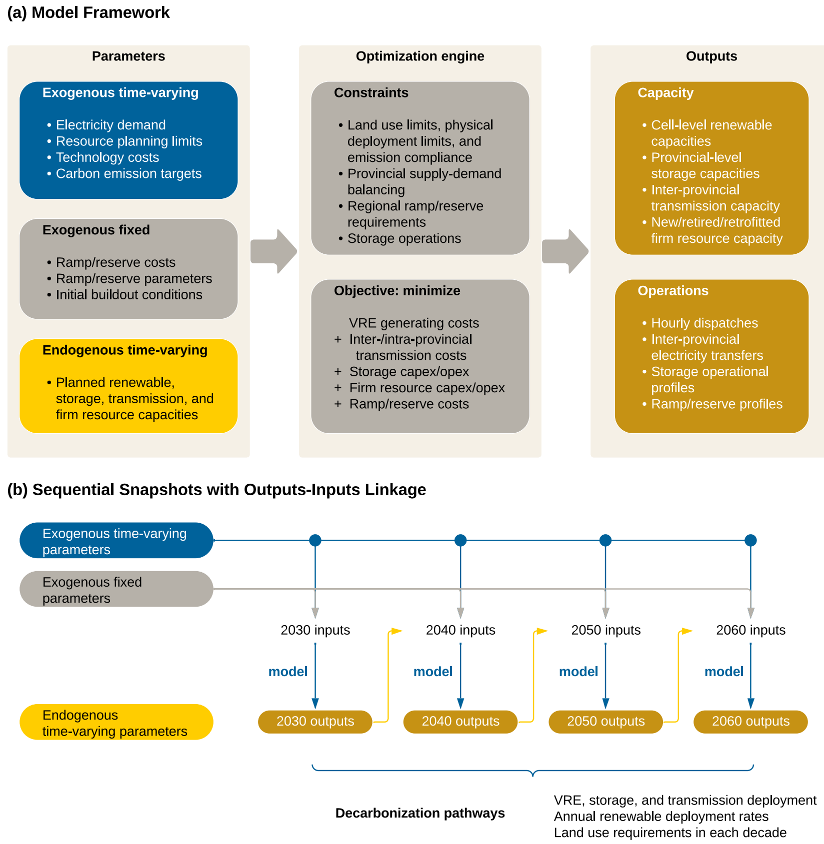

Fig. 1. Modeling framework for decarbonization pathways of China's power sector. The modeling framework in panel (a) includes model parameters, constraints, objectives, and outputs. Panel (b) shows the outputs-inputs linkage between sequential snapshots. Exogenous parameters are fed into each model run, and model outputs of one run are used as inputs for the next run.

Table 1
Model scenarios and sensitivity analysis. Four scenarios are designed around carbon emission targets, CCS availability, and district heating modes. Sensitivity analysis is conducted under the 2 °C baseline scenario, considering renewable cost declines, CCS retrofit costs, and electricity demand increases.

Scenario	2 °C baseline	1.5 °C	2 °C without CCS	2 °C with heat pumps
Demand trajectories	Normal	6%–10% increases relative to the normal trajectory	Normal	Normal with adjustments
Carbon emission trajectories	2 °C	1.5 °C	2 °C	2 °C
CCS availability	2040	2040	No CCS retrofits	2040
District heating mode	Centralized CHP CCS	Centralized CHP CCS	No CHP CCS plants	Heat pumps

run. We explore parametric sensitivities such as investment costs and further model pathways scenarios as shown in Table 1. We find that annual renewable deployment rates will need to increase by 2–3 times recent historical trends to meet emission targets for the power sector. Both regions with lower-quality and higher-quality resources will see onshore wind and distributed solar deployed, while regions with high-quality resources are more sensitive to carbon capture and storage (CCS) availability and district heating modes. Our research highlights the regional heterogeneities over time and provides directions for policy support going forward.

2. Methods

2.1. Power systems modeling procedures

We develop a modeling framework that extends the Renewable Energy Siting and Power-system Optimization (RESPO) model to simulate China's power sector by decade from 2020 to 2060 [35]. The model formulation is provided in Supplementary Note 1. The model optimizes the amount and location of firm resources, variable renewable energy, storage systems, and transmission lines at least cost for each time period. Eleven types of generation technologies are considered: onshore wind, offshore wind, utility-scale solar, distributed solar, coal, gas, coal-CCS, gas-CCS, biomass-CCS (BECCS), hydro, and nuclear. Four types of storage systems (i.e., pumped hydro storage, battery storage, long-duration compressed air energy storage, and long-duration vanadium redox-flow battery), and three types of transmission lines (i.e., spur line, trunk line, and inter-provincial line) are also considered. Spur lines and trunk lines are intra-provincial lines that connect wind and solar to the high-voltage power grid and are factored into renewable deployment costs.

The objective of each simulation is to minimize the annualized capital, maintenance, and operating costs at the national level. Each simulation is subject to a set of planning and operation constraints. Planning constraints reflect initial brownfield conditions for all generation types, land use limits for renewables, physical deployment limits for firm resources, and carbon emission targets for the power sector. We do not impose any limits on VRE deployment rates as in other studies [6], because this study focuses on the implications of deployment patterns. Operation constraints include electricity demands, resource generation limits, transmission operations, reserve requirements, ramping limits, and storage dynamics at the provincial level on an hourly basis for a whole year of operation. Investment decisions, generator dispatch, and inter-provincial electricity transfers are obtained from the simulation outputs for further modeling and analysis. Detailed firm resource planning limits are provided in Supplementary Note 6.

The model is developed in Python and formulated as a linear programming problem that consists of up to 28 million variables and 27 million convex constraints. We use Gurobi v9.5.2 as the optimization solver and select Barrier as the algorithm. Using 32 cores from an Intel Xeon processor and 256 GB of memory, each simulation takes up to 20 h to complete. Given the computational requirements, we do not consider additional plant-level operation constraints (e.g., unit commitment). We advance traditional system planning and operation models with more detailed land use representations and the endogenization of coal and gas retirement and CCS retrofit decisions, consistent with China's recent policy efforts [36].

To uncover realistic pathways for decarbonizing China's power sector, we adapt this model to evolving power system conditions and simulate potential technology lock-in and vintaging effects by sequentially linking the outputs of each decade's simulation myopically. The sequential simulations include exogenous parameters that are obtained from relevant modeling studies and policy documents as input. These parameters can be classified into two main categories: fixed parameters (e.g., technical properties, and operating requirements), and time-varying parameters (e.g., electricity demands, and technology costs). For electricity demand, we adopt projections at the national and provincial levels from an economy-wide decarbonization study [37] and scale up provincial profiles to match annual demand projections [38]. For technology costs, we obtain the 2020 level in the context of the Chinese market and adopt cost decline trajectories for renewables and storage systems [35]. Detailed input assumptions are provided in Supplementary Note 2, 3, and 7.

2.2. Technical and economic assessment of VRE

To conduct a technical assessment of VRE resources, we first obtain hourly simulation records of atmospheric properties including temperature, wind speeds, and solar irradiance in 2015 at a spatial resolution of 0.3125° longitude by 0.25° latitude (approximately 30 by 28 km per grid cell at mid-latitudes) from the Goddard Earth Observing System (GEOS-5). Using the technical parameters of General Electric 2.5-MW and Vestas 8.0-MW turbines, we estimate hourly capacity factors (CF) at 100-m hub height for onshore and offshore wind, respectively. Similarly, using the technical parameters of a fixed-tilt solar module (e.g., 19.9% module conversion efficiency, and 83.2% system performance coefficient) in addition to the GEOS-5 records, we estimate hourly capacity factors at 2 m above the displacement height for both utility-scale and distributed solar. We use the observed records from the Global Solar Atlas to adjust the calculated capacity factors.

To calculate the maximum deployment potential per grid cell, we first calculate the suitable areas per grid cell for renewable deployment based on land use data in 2018 at a 30-m spatial resolution. In this procedure, we consider a wide range of land suitability assumptions, such as preferences for low slopes and altitudes and for certain land types with fewer competing uses. For example, the National Energy Administration and other ministries have issued land use regulations that restrict solar deployment on productive agricultural lands, where we assume only a small fraction (2%–20%) is available for solar, while wind can be deployed on most (80%–100%) of the croplands, grasslands, shrublands, and barren lands. Detailed land use assumptions are provided in our previous work [35].

Then, we assume that each wind turbine requires an area of 10 rotor diameters in the downstream direction and 5 rotor diameters in the crosswind direction to calculate the deployment potential in MW per grid cell for both onshore and offshore wind. For utility-scale solar, we assume each square kilometer area can deploy 161.9 MW of solar and discount the deployment potential by packing factors as a function of latitude. For distributed solar, we first estimate the built-up area per grid cell using NASA's Moderate-Resolution Imaging Spectroradiometer (MODIS) land cover data. Then, we assume 15% of the built-up areas are buildings and 50% of their rooftop areas are available for deploying distributed solar.

We adopt a levelized-cost-of-electricity (LCOE) method to conduct an economic assessment for each 0.3125° by 0.25° grid cell. Total lifetime costs of wind and solar include generation (i.e., capital, maintenance, and operations) and grid connection costs. The generation component is estimated based on the input assumptions of renewable costs considering percentage decreases in each simulation period. The grid connection component refers to the costs of connecting wind and solar resources in each grid cell to the nearest major node on the transmission network. Specifically, spur lines are used to connect cells to substations, and trunk lines are used to further connect substations to the nearest major nodes. In this approach, we simultaneously determine the capacities of spur and trunk lines, and wind and solar capacities in each grid cell. Lastly, we use the total lifetime costs and the estimated capacity factors to calculate cell-level LCOEs. Note that we do not consider distribution network details. A detailed comparison with existing literature on modeling is provided in Supplementary Note 4.

2.3. Scenarios and sensitivity analysis

To evaluate the impacts of parametric and structural uncertainties on decarbonization pathways, four scenarios are designed around (1) policy stringency (2 °C and 1.5 °C), (2) CCS availability, and (3) district heating modes. Previous studies have constructed scenarios around carbon emission targets [4–6,35], renewable capital costs [4–6,35], electricity demand trajectories [4,6], transmission expansions [5,6], operational requirements [5,6,35], land uses [35], hydrogen demand, and electric vehicle charging requirements [5]. We expand the scope of uncertainty analysis by examining the impacts of delaying CCS availability and switching district heating modes on renewable deployment.

For the first two scenarios, we obtain carbon emission targets and corresponding electricity demand projections across decades under both 2 °C and 1.5 °C global average temperature rise targets from an economy-wide decarbonization study [37]. If more stringent carbon emission targets are imposed on the power sector, we assume other sectors will have higher levels of electrification, leading to more electricity demand growth in each decade in the second scenario.

In the third scenario, we delay the CCS availability from 2040 to later periods. Studies on the role of CCS in power systems highlight several deployment challenges attributable to the high marginal costs of carbon abatement, lack of viable business models, and poor carbon source–sink match [39]. Specifically, BECCS, though considered vital [40], has raised concerns about land use changes, fertilizer uses, and upstream emissions from biomass supply chains [41,42]. Therefore, despite recent CCS deployment trends in China, we delay the introduction of CCS to later periods considering potential technical, economic, or political bottlenecks, in the third scenario.

In the last scenario, we switch the district heating mode from centralized CHP plants to heat pumps. Studies on low-carbon district heating systems propose various transitional modes: centralized (e.g., city-scale CHP plants, large-scale heat pumps, large boilers), decentralized (e.g., substation-level heat pumps), or mixed configurations [43]. The urban district heating demand in China was 3.3 billion GJ in 2019 [44] and is projected to decrease to 3.1 billion GJ by 2060 due to building energy efficiency improvements based on forecasts of urban areas, central heating share, and heating demand per unit area [3]. Under the baseline scenario, we assume that an increasing amount of urban district heating demand will be met by centralized CHP CCS plants starting in 2040, given that northern cities in China use primarily CHP as the main heating source with a massive district heating network. The rest of the urban district heating demand will be met by other sources (e.g., conventional CHP plants, fuel boilers, and heat pumps), whose emissions are not directly counted towards the power sector. We scale the CHP capacity in 2019 to estimate the CHP CCS capacity from 2040 to 2060 as exogenous model inputs.

Although heat pump adoption rates are currently still low in urban areas, we explore the possibility of using heat pumps as the dominant strategy to decarbonize the urban district heating demand in northern China [45]. Heat pumps are more likely than resistance-based alternatives due to efficiencies and compatibility with the existing district heating network [45]. In this case, the portion of the heating demand originally met by CHP CCS will need to be met by heat pumps, which increases electricity demands in northern provinces and requires more VRE deployment. To examine this impact, we first obtain monthly average temperatures at the provincial level from the website of the China Meteorological Administration [46] and derive the coefficient of performance (COP) of heat pumps as a function of temperatures. Then, assuming a 1.5 heat-to-power ratio for coal-fired CHP plants [47], we calculate the induced demand increases and update the simulation inputs.

In the sensitivity analysis, we evaluate the impacts of renewable cost declines, CCS retrofit costs, and electricity demand growth. As future electricity demands highly depend on the electrification levels of other sectors, we assume that the electricity demand in each province will increase or decrease by 5% relative to the baseline. In addition, we start with 2020 capital costs from China but assume fewer percentage decreases in renewable cost declines based on technology-specific projections of the conservative category released in 2023 by the U.S. National Renewable Energy Laboratory (NREL) [48]. Detailed scenario differences and sensitivity analysis are provided in Supplementary Note 5.

3. Results

3.1. Temporal limitations of transition pathways

For China to reach carbon neutrality, a significant increase in electricity demand and a sharp reduction in emission rates will require a large increase in renewable energy and transmission infrastructure. Fig. 2 presents the installed VRE and storage capacities and deployment rates nationwide in each scenario. Across all the 2 °C scenarios, total wind capacity increases from 770 GW in the first decade (2020–2030) to 2200–3400 GW in the last decade (2050–2060), and total solar capacity increases from 850 GW to 3400–4800 GW. If CCS is not available, an extra 1100 GW of wind and 990 GW of solar will be needed nationwide relative to the 2 °C baseline, requiring an additional annualized investment of around 1.7 trillion RMB in VRE in the last decade, measured by the 2020 prices. Furthermore, if the electrification process of the district heating network is accelerated, an extra 380 GW of wind will be required, while solar capacity will decrease by 410 GW, demonstrating larger complementarity between heating demands and wind resources.

Despite the seemingly plausible transition pathways, meeting the requirements for renewable capacities can face substantial temporal limitations nationwide. To show this, we calculate the annual average deployment rate per decade based on installed renewable capacities. In Fig. 2, the future required renewable deployment rates across the four scenarios are benchmarked by historical rates. Under the 2 °C target, the combined deployment rate for wind and solar will need to increase from 110 GW/year in the first decade (2020–2030) to 330–420 GW/year in later periods. To reach the 1.5 °C target, we find a significant jump to 180 GW/year in the combined deployment rate in the first decade (2020–2030), followed by a dip to 80 GW/year in the second decade (2030–2040). This dip suggests it is more economical to retrofit existing thermal generators with CCS than to deploy renewables at low capacity factors under the 1.5 °C scenario. Similarly, to reach the storage deployment requirements, the deployment rate of storage across all the 2 °C scenarios will need to increase from 10 GW/year in the first decade (2020–2030) to 50–140 GW/year in later periods.

Renewable energy deployment over the next decade is consistent with recent historical rates, though significantly higher deployment is needed in later decades. The average deployment rates for wind and solar in the past three years (2021–2023) since China's announcement of carbon neutrality goals are 53 and 119 GW/year, respectively [8,49]. The combined renewable deployment rate is 56% higher than the required deployment rate in the first decade (2020–2030) across the 2 °C scenarios. Furthermore, the deployment rates for pumped hydro and battery storage were 7 and 14 GW/year since 2022 [50–52], respectively, twice the requirements under the 2 °C scenarios.

3.2. Subnational barriers to renewable scale-up

China's six geographical grid regions (i.e., East, Central, South, North, Northeast, and Northwest) will experience concentrated deployment efforts in different decades. To show this, we calculate the annual average deployment rate per decade, for each grid region. An annual average rate requirement higher than historical rates at subnational levels can be politically challenging if the deployment entails drastic coal retirement efforts, and economically suboptimal if inter-regional trading mechanisms are lacking.

Fig. 3 shows the annual average deployment rates of wind and solar across regions. We find that all the grid regions will have maximum annual average rates occur in the last decade (2050–2060), ranging from 12–34 GW/year for wind and 23–67 GW/year for solar under the 2 °C baseline. Annual average rates can have slight decreases in the second decade (2030–2040) due to the availability of CCS retrofits but later bounce back. With the option to retrofit existing thermal generators with CCS, high-CF regions deploy more CCS and less VRE for inter-regional transfers. Under the 1.5 °C target, this pattern is more visible: more stringent carbon emission targets will lead to more VRE deployment early on.

Delaying the introduction of CCS to later periods shifts maximum annual VRE deployment rates to early decades. Northwest China, a region with high-CF solar resources, will be mostly impacted by CCS availability, where the annual average deployment rate of solar will need to increase from its current 11 GW/year to 49 GW/year by the end of the last decade (2050–2060) under the 2 °C baseline. But if CCS is not available, this figure will need to jump higher by 57% to 77 GW/year and start a decade earlier.

Combined heat and power (CHP) CCS plants are used to meet district heating demand in northern urban areas during wintertime under the 2 °C baseline. If more heat pumps are deployed to replace the centralized CHP CCS plants, more VRE will be needed in the early decades to manage the induced electricity demand increases. The Three-North regions (Northeast, North, and Northwest China) with the most high-CF wind resources will see maximum annual average deployment rates of wind increase by 5–13 GW/year (36%–46%), followed by East China's jump of 5 GW/year (10%). In addition, storage capacity will decrease by 120 GW, while inter-provincial transmission capacity will increase by 240 GW by the end of 2060 to accommodate the structural changes in renewable deployment relative to the CHP CCS case.

3.3. Shifts in geographical distributions of renewables and transmission infrastructure

Grid regions have different deployment patterns over time, influenced by their high and low capacity factors driven by cost. Fig. 4 displays the cell-level VRE capacity additions in each decade under the 2 °C baseline scenario. Onshore wind starts from both high-CF regions (i.e., North China and Northeast China) and low-CF regions (i.e., East China) in the first decade (2020–2030). For onshore wind, high CF

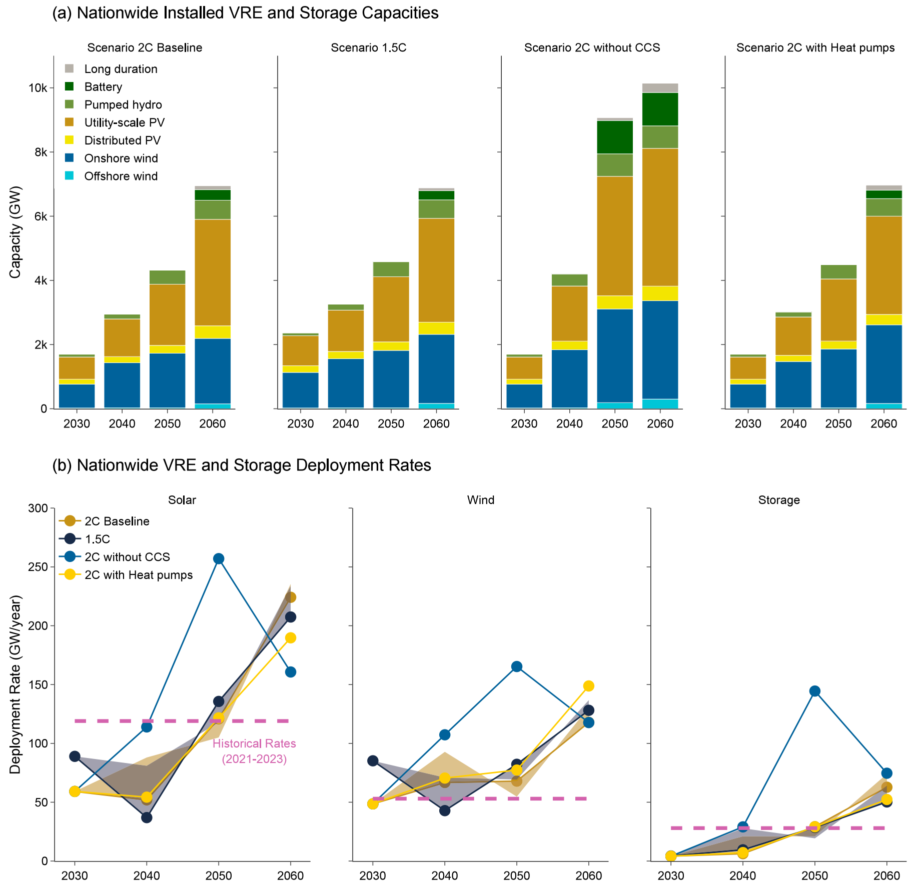

Fig. 2. Nationwide installed VRE and storage capacities and deployment rates across four scenarios. Note that we do not show capacities for coal, gas, nuclear, hydro, and biomass. In panel (b), the red dashed lines represent historical average deployment rates calculated based on capacity additions from 2021–2023. The banded area shows the range of deployment rates in the sensitivity analysis. The calculations incorporate new builds to replace retiring assets. We assume all existing VRE capacity in 2020 will retire by 2045. All existing storage capacity in 2020 will retire by 2060 for pumped hydro and by 2035 for battery systems.

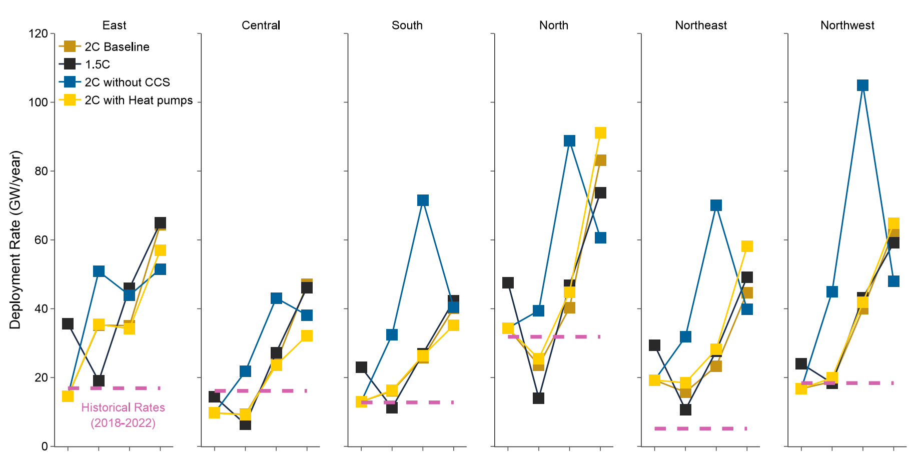

Fig. 3. Subnational VRE deployment rates by grid region across four scenarios. Annual VRE deployment rates in each decade are calculated based on regional-level installed capacities. Red dashed lines indicate each region's average deployment rates from 2018–2022.

implies lower generation costs and low CF generally implies lower transmission requirements. Therefore, both regions are preferred from the least system cost perspective. This trend continues in later periods, primarily driven by the coal retirement in both high-CF and low-CF regions. In comparison, offshore wind only sees limited deployment in the first decade (2020–2030) due to its assumed high capital costs relative to other renewable technology types. However, cost declines will lead to a significant ramp-up in high-CF (i.e., coast) regions in the last decade (2050–2060).

Utility-scale solar starts predominantly from high-CF regions (i.e., North China and Northwest China) in the first decade (2020–2030), then spreads to low-CF regions by the second decade (2030–2040)

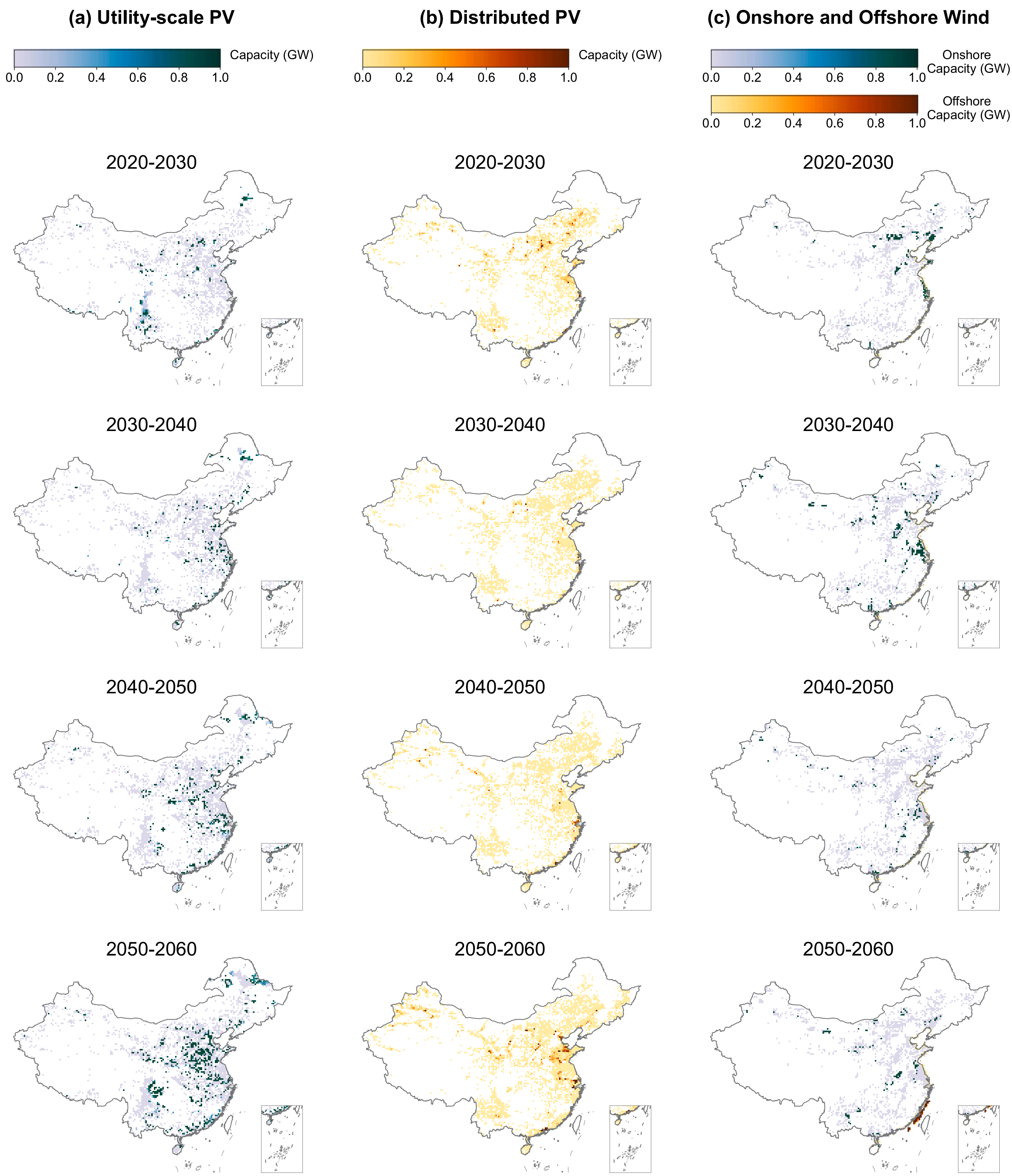

Fig. 4. Cell-level VRE capacity additions under the 2 °C Baseline scenario. Each grid cell corresponds to an area of approximately 30 by 28 km at mid-latitudes. The results are categorized by VRE technology types, including utility-scale solar, distributed solar, onshore wind, and offshore wind.

as high-quality resources are exploited. In contrast, distributed solar follows a similar pattern as onshore wind: it is built in both high-CF and low-CF regions throughout the pathway to neutrality. Overall, we observe fewer regional heterogeneities in solar deployment than wind, largely attributable to resource endowment across different grid regions.

Fig. 5 shows the geographical distributions of the newly added transmission capacities in each decade. The first decade (2020–2030)

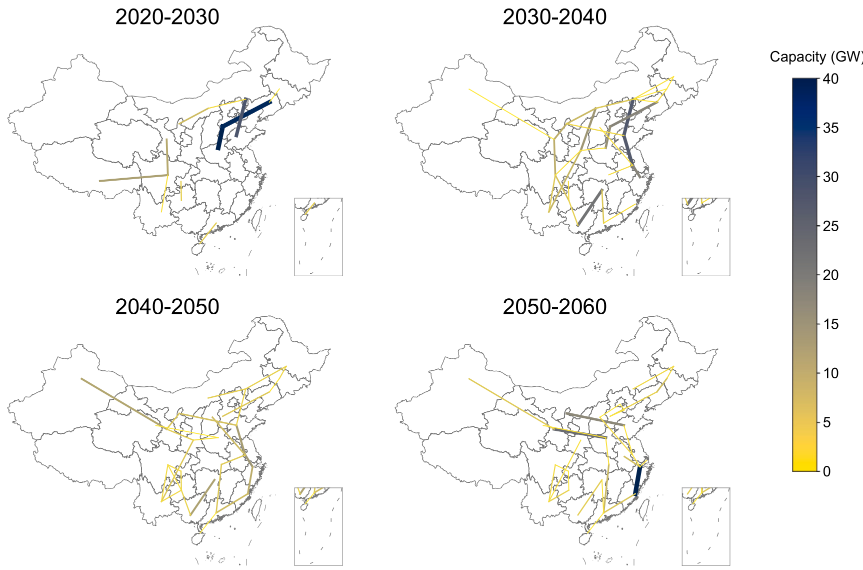

Fig. 5. Newly added inter-provincial transmission capacities under the 2 °C Baseline scenario.

sees transmission lines built to connect high- and low-CF regions and accommodate mostly onshore wind deployment. In later decades, we observe new transmission lines both between high- and low-CF regions, and between low-CF regions. The average utilization rates of transmission lines between low-CF regions increase from 31% to 62% in the last decade (2050–2060), and the utilization rates from high-CF to low-CF regions increase from 27% to 50% in the last decade (2050–2060). These patterns suggest that the role of transmission lines expands from managing power transfers from high-CF to low-CF regions to balancing load between regions with diverse renewable supplies.

3.4. Land use constraints and manufacturing limitations

Land use constraints impact China's renewable energy pathways. Solar deployment, including both utility-scale and distributed, has an average usage of 16%, 21%–27%, 24%–40%, and 35%–43% of suitable land from 2030 to 2060 across all the 2 °C scenarios, based on land use suitability assumptions described in [35]. Thus, eventually at least one-third of the land that is currently suitable for solar deploys solar. East China and South China will be the most impacted regions in the first decade (2020–2030), where the highest provincial solar land use shares can exceed 40%. In the last decade (2050–2060), East China and North China will exploit the majority (21%–82%) of suitable land for solar. In contrast, wind deployment, including both onshore and offshore, impacts land use to a lesser extent than solar as it can be deployed on more types of land (including agricultural land). The last decade (2050–2060) will see 22%–45% of suitable land exhausted for wind in EC.

To consider manufacturing limitations on capacity expansion decisions, we analyze deployment requirements against current and projected manufacturing capacities. On the supply slide, manufacturing capacities for main wind power components and solar modules are projected to reach 89 GW/year and 900 GW/year [53] by the end of 2024, respectively. China's solar manufacturing capacity should be sufficient for handling domestic demands, though we do not model expected export demands. However, wind manufacturing still falls short of the wind deployment requirements by around 50% in the long term. For storage systems, China's manufacturing capacity for lithium-ion batteries is projected to reach 4.65 TWh by 2030 [54], sufficient to meet long-term storage demands. Lastly, considering that the past decade has seen ultra-high-voltage lines expand at a rate of approximately 16–20 GW/year [55], meeting the near-term transmission expansion requirements should be plausible.

Lastly, we conduct a sensitivity analysis on the uncertainty from renewable cost declines, CCS retrofit costs, and electricity demand growth. Under various electricity demand growth rates, our findings about the geographical distribution and regional impacts remain robust. The combined deployment rate for wind and solar ranges from 85–130 GW/year in the first decade (2020–2030) to 320–380 GW/year in the last decade (2050–2060). In comparison, more conservative renewable costs lead to changes in the relative shares of renewable technologies in the capacity mix—in particular, distributed solar sees higher installations in later decades. Relatedly, maximum annual average rates of solar will have fewer regional variations. Increasing CCS retrofit costs will incur more deployment requirements for solar than for wind across all decades, and the impacts will be mostly concentrated in regions with high-CF solar resources (i.e., North China and Northwest China).

4. Discussion

4.1. Model implications

Our modeled output reveals important temporal differences compared to relevant power systems modeling studies [4–6,35]. We see 100–450 GW of coal and gas CCS plants, significantly higher than [6] as we allow the model to retrofit existing thermal generators with CCS. Gas CCS plants will be built in the last decade (2050–2060) primarily to meet system planning reserve requirements, which might change as technology matures and alternative cost futures are realized. Storage and transmission lines have similar total installed capacities measured in GW but can complement each other depending on input cost assumptions [5,6]. Low-CF regions will see more onshore wind and distributed solar deployment in the early decades than in other studies, which can be attributed to the higher transmission requirements of high-CF regions as we endogenize the costs of expanding intra-regional lines. Furthermore, CCS availability and district heating modes will primarily raise the deployment requirements in high-CF regions. This suggests it will be critical for the national government and specific regions that will bear the brunt of this increase to incorporate these impacts into VRE planning processes.

Most relevant studies focus on the end-state of a net-zero electric power system, and few have analyzed the temporal limitations with annual deployment rates at national and subnational levels. This study highlights the heterogeneity in deployment requirements across regions. We find that most grid regions will see the highest annual average deployment rates in the last decade (2050–2060), but certain regions will need to increase deployment rates 3–4 times more than others (e.g., wind in East China, and solar in North China and Northwest China). These variations imply potential deployment barriers. First, regions with less available land for renewables will face land use pressure earlier than others and thus should prioritize efforts for comprehensive land use planning pilots. Second, regions with high coal and gas capacity will need to plan for the societal risks associated with reduced thermal runtime and a displaced workforce. To make the transition more equitable, the national government should identify strategies tailored to local contexts that also prioritize renewable deployment, and more broadly, provide transition assistance programs such as workforce development in impacted regions.

4.2. Model limitations

This study has several limitations. Operational and market frictions are widespread in China and can impose restrictions on inter-provincial transfers and VRE integration [56]. Our model assumes that provinces can trade power freely with any other connected province up to total transmission interconnection capacities. Electricity markets will need to further evolve to accommodate the increasing renewable penetration and address the market inefficiencies. We do not explicitly model technological learning effects, which are key to continued cost declines [26]. Therefore, our model could underestimate the necessary early deployments of certain technologies such as battery storage and offshore wind. Lastly, our renewable resource assessment assumes static land use policies, but local governments may provide more clarity on land use criteria and impose additional land use regulations, which will disproportionately impact resource availability [35]. We believe these are promising directions for future research on decarbonization pathways.

4.3. Policy implications

The evolution of regional heterogeneities in VRE depends on uncertain factors such as renewable cost declines, demand growth, technological advancements, and interactions with non-power sectors. On the one hand, regions that adopt renewable deployment incentives can develop policies and regulations that are transferrable to other regions in later periods [9,11,57]. On the other hand, regions should take differentiated strategies according to local contexts, as shown in Fig. 6 [58–60]. Regions with large-scale VRE deployment should focus on establishing manufacturing hubs and creating innovative financial incentives. Regions with high fossil fuel retirement should mitigate the resulting socioeconomic impacts and implement workforce placement programs. Lastly, regions with smaller VRE or CCS capacity should identify economic diversification opportunities to support other clean energy industries.

5. Conclusion

Temporal and subnational limitations of variable renewable energy (VRE) deployment and transmission expansion in different regions can be the bottleneck to a country's decarbonization progress, despite national planning efforts. Furthermore, technology uncertainty and policy stringency can further exacerbate temporal limitations at subnational levels, rendering cost-efficient pathways infeasible. In this work, we conduct a pathway analysis with high spatial resolution, and examine where and when temporal limitations will be most substantial, thus highlighting the policy efforts required locally.

Our study shows that China will need to deploy wind and solar at increasing rates—rising from 110 GW/year in the first decade (2020–2030) to 330–420 GW/year in the last decade (2050–2060), in addition to supporting infrastructure including storage and transmission lines, to achieve the 2 °C global average temperature rise target. At subnational levels, 45% of the provinces are deploying slower than the requirements in the first decade, and this number will increase to 100% in the last decade. If a more stringent climate target is imposed or if carbon capture and storage (CCS) is not available, regions with high-quality resources will see more concentrated VRE deployment, which occurs sooner.

We quantify the acute deployment pressure at the regional level under uncertain factors and identify region-specific policy priorities for local governments. Through this analytical framework, we can combine technical, operational, and land use details as well as political economy challenges that can occur while scaling up renewable deployment. The implications for both power system modelers and policymakers are that acute temporal impacts and subnational limitations should be examined to ensure the feasibility of transitional pathways to net zero.

CRediT authorship contribution statement

Zhenhua Zhang: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Ziheng Zhu:** Writing – review & editing, Visualization, Software, Methodology, Data curation. **Jessica A. Gordon:** Writing – review & editing. **Xi Lu:** Writing – review & editing, Data curation. **Da Zhang:** Writing – review & editing, Supervision, Funding acquisition. **Michael R. Davidson:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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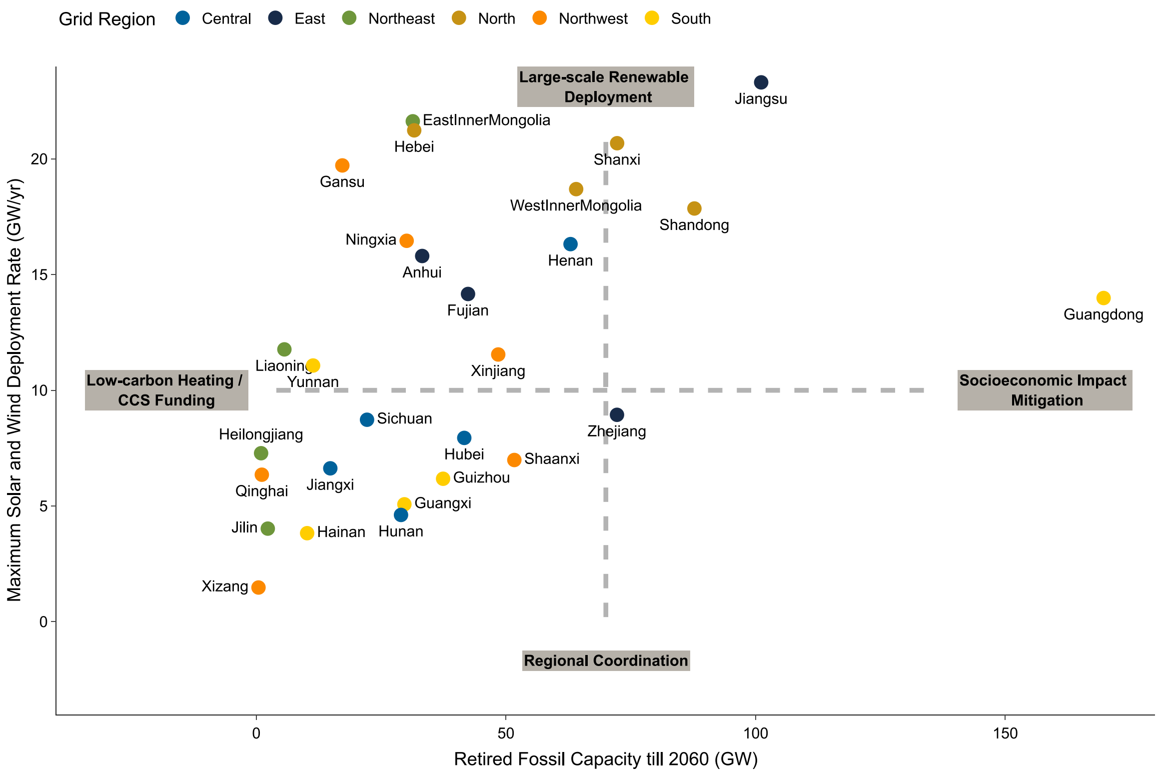
Supplementary data

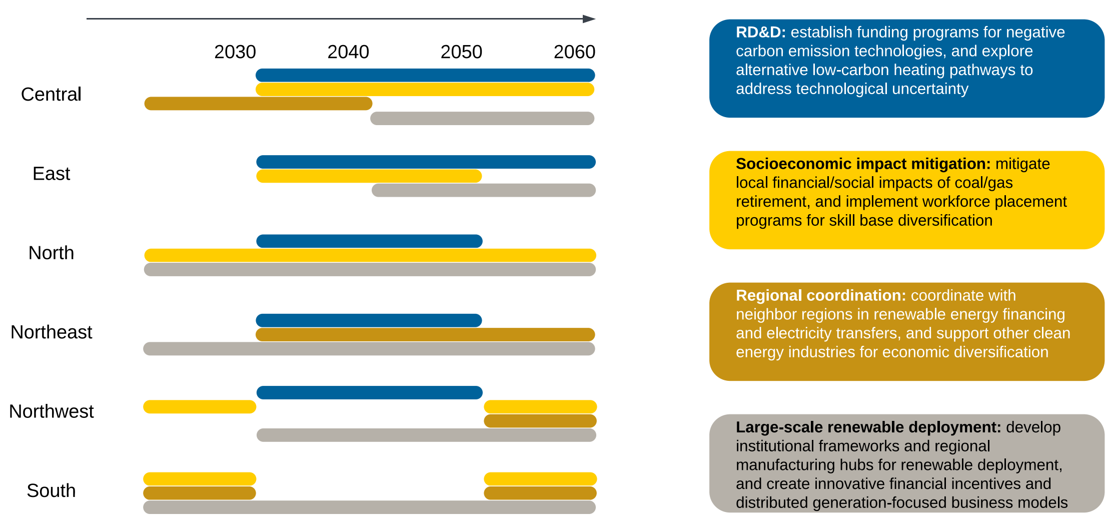
Supplementary material related to this article can be found online at https://doi.org/10.1016/j.adapen.2025.100238.

Data availability

All source data, code, modeling inputs, and generated outputs from this study are available on Zenodo at https://doi.org/10.5281/zenodo.14907700. The code used for this study, including analysis scripts, is available on GitHub at https://github.com/Power-Lab/AdvAppliedEnergy_Pathways_2025.



(a) Provincial policy priorities



(b) Regional policy priorities over time

Fig. 6. Regional policy focuses in the face of future renewable deployment and fossil retirement requirements. In panel (a), icon colors represent the grid region for each province. Municipalities are not shown here due to their small numerical values. Policy focuses for each group are explained in detail in panel (b).

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